

HelixVPN — Master Specification Index (the full document tree)

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Revision: 5 **Last modified:** 2026-07-04T12:00:00Z **Status:** active — the navigation + generation blueprint for the full HelixVPN technical specification **Rev 5 (2026-07-04 hardening pass):** Independent deep gap-analysis + direct hardening executed across **every** volume (V0-V10, ~140 nano-detail docs + the 16 top-level final/ chapters + all 5 04_VPN_CLD source docs) via 16 parallel review-and-fix passes, closing the previously-owed adversarial reviews for V0/V1/V6/V7/V8 (V2/V3/V4/V5/V10 had already passed review pre-Rev-5; this pass re-verified them independently rather than trusting the "GO" label, and found + fixed several real defects even in the already-reviewed volumes — see below). Real cross-document defects found and fixed (not padding): (1) a security-relevant PKI bug — CA rotation had conflated *scheduled* rotation with *compromise-triggered* rotation, meaning a compromised issuing CA would have stayed trusted for up to 24h (v03-control-plane/svc-pki.md, now split into two distinct paths); (2) a real Rust contradiction — the FFI boundary claimed both `catch_unwind` panic recovery *and* workspace-wide `panic=abort`, which are mutually exclusive (v04-client/helix-core-rust.md, now an honest corrected contract); (3) a missing QUIC anti-amplification limit on the public MASQUE edge listener — a real DDoS-vector gap (v02-data-plane/transport-masque-quic.md); (4) a dangling reconcile-reliability gap — no schema existed to prevent a silently-dropped reconcile trigger on a crash between the Postgres commit and the Redis XADD (v03-control-plane/data-model-ddl.md, now an outbox table + sweeper); (5) rate-limiting/DDoS resilience and a resource-sizing/cost model were genuinely absent anywhere in the 131-file corpus — both added (05-repo-layout-tooling-and-helix-ecosystem.md, v06-deploy/ha-and-multiregion.md); (6) R5 (Phase 1-3 WBS task/subtask tier asymmetry, tracked in

REFINEMENT_NOTES.md) resolved — the already-existing v07-execution/subtask-deepening-p1/p2/p3.md docs were orphaned (nothing in 07/08/09-*.wbs.md pointed to them); now cross-referenced; (7) D-PKI-CA-TIER and D-0D-1 (both noted below as open in Rev 3/4) are **operator-confirmed** per v00-meta/decision-register.md Rev 2 — the Rev-3/4 notes below are superseded, kept for history; (8) a numeric contradiction between SPECIFICATION.md §8.2 ("< 30s" failover) and 08-phase2-parity-wbs.md's measured **P2-SLO3** ("< 3s") reconciled in SPECIFICATION.md Rev 2 (the WBS's tighter, instrumented figure wins). Every touched file carries its own Revision bump recording its specific changes; consult each file's header for the full per-doc changelog. §11.4.168 Mermaid-pandoc-render tooling gap (item (b) below) remains open — it is a build-tooling fix, not a markdown-content fix, and stays out of this pass's scope. **Rev 4:** Re-marked the generated vNN-* nano-detail rows [GEN]→[DONE] across Volumes 0-10 (the expansion docs now exist on disk with synced siblings); added the [DONE] status to the legend. The [P1] overview rows stay [P1]; [RES] preserved (e.g. [DONE][RES]). **Rev 3:** All 11 volumes (V0-V10) expanded — generation COMPLETE (~140 nano-detail docs, all with synced HTML/PDF siblings). Marked the wave order done; recorded the tracked pre-tag quality items (owed volume reviews, §11.4.168 Mermaid-render, V9 dossiers, D-PKI-CA-TIER reversibility). **Rev 2:** Added **Volume 10 — Design System (OpenDesign)** as a decoupled reusable submodule (vasic-digital/helix_design) per operator mandate 2026-06-25 (§11.4.162); repointed the V4 design row to the client-integration view; recorded the resumed wave order.

This index decomposes the HelixVPN specification into **volumes of nano-detail documents** (dozens of documents, thousands of pages target). SPECIFICATION.md is the architectural spine; this index is the complete map. Pass-1 docs (the NN-*.md topical set) are the **volume overviews**; each volume's vNN-<area>/ subdirectory holds the deep per-topic documents generated in expansion waves. No false results, no bluff: every deep doc cites sources [04_ARCH \$N]/[research-<angle>] and marks anything unproven UNVERIFIED (§11.4.6).

Status legend

- **[P1]** authored in pass 1 (exists) — serves as the volume overview, to be deepened.
 - **[GEN]** to be generated in an expansion wave (nano-detail).
 - **[DONE]** generated + on disk + HTML/PDF siblings synced (an expansion-wave doc that has landed).
 - **[RES]** research-backed (cited corpus in 11-deep-research-appendix.md).
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Volume 0 — Spine, meta & governance

Doc	Status	Scope
SPECIFICATION.md	[P1]	Architectural spine: roles, principles, one-screen architecture, roadmap, decision register, glossary
MASTER_INDEX.md	[P1]	This file — the full document tree
99-source-coverage-ledger.md	[P1]	Source-coverage proof (16 docs → where absorbed), gaps
REFINEMENT_NOTES.md	[P1]	Pass-1→pass-N punch-list
v00-meta/glossary.md	[DONE]	Full glossary (every term, acronym, protocol, RFC)
v00-meta/decision-register.md	[DONE]	D1-D8 expanded + the gate that resolves each + reversal criteria
v00-meta/requirements-traceability.md	[DONE]	Requirement → component → test cross-reference matrix

Volume 1 — Product & requirements

Doc	Status	Scope
00-product-scope-and-principles.md	[P1]	Overview: product def, personas, roles, scope, principles, parity matrix, decisions
v01-product/product-vision-and-positioning.md	[DONE]	Vision, market, prior-art deep-dive, differentiators, non-goals
v01-product/personas-and-roles.md	[DONE]	Each persona: jobs-to-be-done, journeys, pain points, success
v01-product/functional-requirements.md	[DONE]	Every FR, numbered (HVPN-FR-NNN), with acceptance
v01-product/nonfunctional-requirements.md	[DONE]	Perf/scale/availability/privacy/security SLOs (HVPN-NFR-NNN)
v01-product/use-cases-and-journeys.md	[DONE]	End-to-end use cases with sequence diagrams
v01-product/success-metrics.md	[DONE]	KPIs, telemetry-derived (counts only, no flows)

Volume 2 — Data plane (Rust)

Doc	Status	Scope
01-data-plane.md	[P1]	Overview: layering, transports, routing, multihop
v02-data-plane/ wireguard-core.md	[DONE] [RES]	boringtun vs kernel WG per platform, Noise IK, AllowedIPs, key mgmt, MTU
v02-data-plane/ transport-trait.md	[DONE]	The Transport trait — full Rust signature, invariants, lifecycle, errors
v02-data-plane/ transport-plain- udp.md	[DONE]	Plain-UDP transport: impl, MTU 1420, fast path
v02-data-plane/ transport-masque- quic.md	[DONE] [RES]	MASQUE CONNECT-UDP (RFC 9298/9297/9221), quinn+h3, hand-rolled framing, DPI camouflage
v02-data-plane/ transport- shadowsocks.md	[DONE] [RES]	Shadowsocks-wrap (AEAD-TCP), shadowsocks-rust
v02-data-plane/ transport-udp-over- tcp.md	[DONE]	UoT last-resort, udp2tcp framing
v02-data-plane/ transport-lwo.md	[DONE]	Lightweight obfuscation (keyed WG-header obfs + padding)
v02-data-plane/ transport- selection-ladder.md	[DONE] [RES]	Auto-escalation, per-network memory, regional priors, handshake-failure events
v02-data-plane/ obfuscation-and- dpi.md	[DONE] [RES]	DPI/censorship landscape (GFW probing, SNI, UDP block), which transport survives what
v02-data-plane/ routing-and- addressing.md	[DONE]	ULA /48, 4via6 overlapping CIDR, AllowedIPs + nftables/eBPF verdict maps
v02-data-plane/ multihop.md	[DONE]	Nested WireGuard, per-hop keys, entry/exit jurisdiction
v02-data-plane/ daita.md	[DONE] [RES]	DAITA via maybenot: packet sizing, cover traffic, timing, overhead
v02-data-plane/ orchestrator-and- state.md	[DONE]	helix-core loops, tokio::broadcast status enum, reconnection state machine

Volume 3 — Control plane (Go)

Doc	Status	Scope
02-control-plane.md	[P1]	Overview: modular monolith, data model, coordinator, events, API
v03-control-plane/architecture-and-wiring.md	[DONE]	Package boundaries, interface/event wiring rules, no-cross-store-import
v03-control-plane/svc-identity.md	[DONE]	Tenants/users/SSO-OIDC/anon-device-token; full API + state
v03-control-plane/svc-registry.md	[DONE]	Devices/connectors/prefixes/overlay-IP registration
v03-control-plane/svc-ipam.md	[DONE]	Overlay pool allocation, ULA / 48, host allocation, 4via6 mapping
v03-control-plane/svc-pki.md	[DONE] [RES]	Tenant CA, short-lived mTLS device certs, rotation, revocation, PQ material
v03-control-plane/svc-policy.md	[DONE]	ACL model, compiler algorithm → AllowedIPs + verdict maps, fail-closed
v03-control-plane/svc-coordinator.md	[DONE]	In-mem topology graph, minimal-delta computation, <1s SLO
v03-control-plane/svc-events.md	[DONE] [RES]	Redis Streams, envelope, taxonomy, consumer groups, XAUTOCLAIM DLQ
v03-control-plane/svc-telemetry.md	[DONE]	Counters/health/audit (no traffic logs), Prometheus metrics
v03-control-plane/svc-api.md	[DONE] [RES]	Gin REST routes, WS/SSE, Connect-RPC, authz, OpenAPI
v03-control-plane/data-model-ddl.md	[DONE]	Full Postgres DDL, per-table, RLS policies, non-superuser role, migrations
v03-control-plane/protobuf-spec.md	[DONE]	Complete .proto (helix.coordinator.v1), every message + field + semantics
v03-control-plane/reconciliation-flow.md	[DONE]	Event→delta→enforce end-to-end, sequence diagrams, idempotency

Volume 4 — Clients (Rust core + Flutter UI + shims)

Doc	Status	Scope
03-client-core-and-ui.md	[P1]	Overview: shared-codebase strategy, FFI, UI, shims
v04-client/helix-core-rust.md	[DONE]	Crate layout, reconciler, kill-switch + DNS state machine, memory/size strategy
v04-client/ffi-surface.md	[DONE] [RES]	flutter_rust_bridge v2 / UniFFI, start/stop/status_stream, TunnelStatus, threading
v04-client/helix-ui-flutter.md	[DONE] [RES]	Melos monorepo, 3 flavors via runHelixApp, capability gating
v04-client/design-system.md	[DONE]	Client-integration view only — how helix-ui consumes helix_design; the full design system is Volume 10 (operator mandate 2026-06-25)
v04-client/state-management.md	[DONE]	Riverpod providers, status-stream→UI pure-function, Console WS/SSE folding
v04-client/shim-apple.md	[DONE] [RES]	iOS/macOS NEPacketTunnelProvider, the memory ceiling, Swift↔Rust
v04-client/shim-android.md	[DONE] [RES]	VpnService + JNI, builder/protect/fd handoff, background-kill
v04-client/shim-windows.md	[DONE]	wireguard-nt/wintun + privileged service, named-pipe IPC, WFP split-tunnel
v04-client/shim-linux.md	[DONE]	kernel WG/tun, systemd integration
v04-client/shim-harmonyos.md	[DONE] [RES]	OpenHarmony flutter fork, Network Kit VPN ability, ArkTS→NAPI→.so
v04-client/shim-aurora.md	[DONE] [RES]	omprussia flutter, Qt/C++ tun shim, signed RPM
v04-client/web-console.md	[DONE]	Console-only build, no core ffi, optional WASM MASQUE proxy caveat

Volume 5 — Security & privacy

Doc	Status	Scope
04-security-privacy-pki.md	[P1]	Overview: zero-trust, identity, PKI, no-logging, PQ
v05-security/threat-model.md	[DONE]	STRIDE/LINDDUN, attacker classes, trust boundaries, mitigations
v05-security/zero-trust-and-default-deny.md	[DONE]	Policy enforcement points, need-to-know peer filtering
v05-security/identity-and-enrollment.md	[DONE]	OIDC + anon device tokens, device keygen, key-never-leaves
v05-security/pki-and-certs.md	[DONE] [RES]	Cert lifecycle, short-lived mTLS, CA hierarchy, rotation, revocation <1s
v05-security/no-logging-as-code.md	[DONE]	CI schema-lint, ephemeral Redis presence, audit-only-control-actions
v05-security/kill-switch-and-dns-leak.md	[DONE]	Per-OS firewall state machine, DNS-forced-through-tunnel
v05-security/post-quantum.md	[DONE] [RES]	ML-KEM/FIPS-203 PSK, hybrid-never-PQ-only, Rosenpass evaluation
v05-security/audit-and-compliance.md	[DONE]	audit_events, control-action audit, compliance posture

Volume 6 — Deployment, tooling & operations

Doc	Status	Scope
05-repo-layout-tooling-and-helix-ecosystem.md	[P1]	Overview: repo layout, codegen, deploy, ecosystem
v06-deploy/repo-layout-and-decoupling.md	[DONE]	Monorepo + planned vasic-digital component repos, §11.4.28/.29/.74
v06-deploy/codegen-pipeline.md	[DONE]	buf → Go/Dart/Rust, OpenAPI → Dart/TS, zero-drift
v06-deploy/helixvpncctl.md	[DONE]	Cobra CLI: init/keys/enroll-token/policy/revoke — full command spec

Doc	Status	Scope
v06-deploy/podman-quadlets.md	[DONE] [RES]	Rootless quadlet units, NET_ADMIN, :443/udp, one-pod, read-only rootfs
v06-deploy/docker-compose.md	[DONE] [RES]	Equivalent Docker Compose stack
v06-deploy/kubernetes.md	[DONE] [RES]	Deployment/StatefulSet/Service/NetworkPolicy manifests
v06-deploy/ha-and-multiregion.md	[DONE]	Stateless coordinators, Patroni PG, NATS JetStream, anycast/geoDNS
v06-deploy/observability.md	[DONE]	Prometheus/Grafana-as-code, convergence + event-lag SLOs
v06-deploy/disaster-recovery.md	[DONE]	RTO/RPO budget, backup, region-failover runbook (closes ledger gap G1)
v06-deploy/helix-ecosystem-integration.md	[DONE]	containers/helix_qa/challenges/docs_chain/security/vision_engine wiring
v06-deploy/remote-testing-infra.md	[DONE]	nezha.local heavy-testing node, containers-submodule distribution (PARKED until greenlit)

Volume 7 — Phase execution (work breakdown → tasks → subtasks)

Doc	Status	Scope
06-phase0-spike-wbs.md	[P1]	Phase 0 WBS (HVPN-P0-NNN), gates G1-G6, milestones S0-S8
07-phase1-mvp-wbs.md	[P1]	Phase 1 MVP WBS (HVPN-P1-NNN), DoD + SLOs
08-phase2-parity-wbs.md	[P1]	Phase 2 WBS (HVPN-P2-NNN)
09-phase3-reach-wbs.md	[P1]	Phase 3 WBS (HVPN-P3-NNN)
v07-execution/workable-items-model.md	[DONE]	§11.4.93 SQLite mapping: every task/subtask → DB row schema + docs_chain sync
v07-execution/dependency-graph.md	[DONE]	Cross-phase dependency DAG, critical path

Doc	Status	Scope
v07-execution/ subtask- deepening-p1.md	[DONE]	Phase 1 epic→task→subtask deepening (closes R5)
v07-execution/ subtask- deepening-p2.md	[DONE]	Phase 2 subtask deepening
v07-execution/ subtask- deepening-p3.md	[DONE]	Phase 3 subtask deepening

Volume 8 — Testing & QA (per §11.4.169 mandatory test types)

Doc	Status	Scope
10-testing- acceptance-and- qa.md	[P1]	Overview: all test types, helix_qa/ challenges, acceptance gates
v08-testing/ unit.md ... v08- testing/ benchmarking.md	[DONE]	One deep doc per §11.4.169 type: unit, integration, e2e, full- automation, challenges, helixqa, ddos, security, stress-chaos, concurrency, race-deadlock, memory, benchmarking — harness, fixtures, evidence, acceptance
v08-testing/ coverage-ledger- schema.md	[DONE]	feature × test-type × evidence- state ledger
v08-testing/test- rig.md	[DONE] [RES]	netns + nftables-DPI + tc-netem rig, iperf3 bars, leak tests

Volume 9 — Research appendix (cited)

Doc	Status	Scope
11-deep- research- appendix.md	[P1] [RES]	Consolidated cited research (10 angles); being rewritten from the corpus
v09-research/ research- <angle>.md ×10	[DONE] [RES]	Per-angle full research dossier (wireguard, masque, hysteria2, mullvad, flutter_ffi, ios_android, go_cp, podman_k8s, pki_pq_nat, daita_test)

Volume 10 — Design System (OpenDesign) — vasic-digital/ helix_design (decoupled reusable submodule)

Operator mandate (2026-06-25, §11.4.162). OpenDesign is the **mandatory** design-and-refinement system for **every** HelixVPN user-facing application (Client, Console, Connector) on **all 8 platforms**. The design system is a **fully decoupled, reusable submodule** (vasic-digital/helix_design, snake_case flat per §11.4.28/.29/.74) incorporable by any future app. This volume documents — in depth — the whole design system: every screen, every reusable component, light+dark color schemes, typography, and **all** design/UI/UX resources emitted in **every consumable form** (JSON / CSS / Dart / Swift / Kotlin / ArkTS / C-Qt) for **direct** incorporation. Every component ships light+dark variants; elements **MUST NOT** overlap/overlay labels; all UI changes carry visual-regression coverage (§11.4.162). **NO** ad-hoc CSS / one-off tools — OpenDesign tokens/themes only; missing patterns extend OpenDesign upstream per §11.4.74.

Doc	Status	Scope
v10-design/00-overview-and-submodule.md	[DONE]	helix_design as a decoupled submodule: purpose, repo layout, semver, consumption model, how ANY app incorporates it (§11.4.28/.74/.162), upstreams/install_upstreams, decoupling invariants (no project-specific context)
v10-design/opendesign-foundation.md	[DONE] [RES]	OpenDesign integration: install, theme/token engine, how it drives palette/type/spacing/component tokens; extend-upstream policy (§11.4.74); honest boundary vs functional/allly testing (§11.4.162)
v10-design/design-tokens.md	[DONE]	Canonical token taxonomy + source-of-truth schema: color/type/spacing/radius/elevation/motion/z-index/breakpoints; naming, tiers (primitive→semantic→component), theming model
v10-design/color-system.md	[DONE]	Full brand palette light+dark , connection-state palette (status colors), semantic mapping, WCAG

Doc	Status	Scope
		contrast proofs, no-overlap/no-overlay rule (§11.4.162)
v10-design/typography-iconography-motion.md	[DONE]	Type ramp + fonts (per-platform), icon set + asset forms, motion/animation tokens & curves
v10-design/token-export-pipeline.md	[DONE] [RES]	Emit tokens in all forms — JSON (Style-Dictionary-class) → CSS vars, Dart ThemeData, SwiftUI, Compose, ArkTS, C/Qt; build pipeline, drift gate, §11.4.65/.168 export + visual validation
v10-design/component-library.md	[DONE]	Every reusable component (ConnectButton, StatusChip, ExitPicker, ShieldToggle, ServerList, PeerCard, TopologyGraph, dialogs, forms, nav) — anatomy, states, variants, tokens, a11y, light+dark
v10-design/screens-client.md	[DONE]	All Client app screens (onboarding, connect/home, exits, multihop, shields, account, settings, errors) — full specs + wireframes, responsive phone/tablet/desktop/TV
v10-design/screens-console.md	[DONE]	All admin Console screens (devices, policy editor, users/SSO, audit, topology, billing) — full specs + wireframes
v10-design/screens-connector.md	[DONE]	All Connector appliance screens (enroll, advertised routes, conflicts, health) — full specs
v10-design/ux-flows-and-interaction.md	[DONE]	End-to-end UX flows, navigation/IA, interaction patterns, motion, loading/empty/error states, i18n, full accessibility spec
v10-design/platform-adaptation.md	[DONE] [RES]	Per-platform UI adaptation (Material/Cupertino/desktop/TV-leanback/HarmonyOS/Aurora) + capability gating, OpenDesign theme mapping per platform
v10-design/visual-regression-and-qa.md	[DONE]	§11.4.162 visual-regression suite, golden screenshots, token-drift tests, a11y/contrast assertions, §11.4.168 exported-doc visual validation
v10-design/assets-and-deliverables.md	[DONE]	All design/UI/UX deliverables in all forms ready for direct incorporation: token bundles, per-platform theme packages, icon/

Doc	Status	Scope
		illustration asset forms, component packages, design-source — + the per-app pull recipe

Generation plan (autonomous, batched, rate-limit-aware)

Documents are generated in **expansion waves** of 3–4 parallel subagents (§11.4.103), each reading the volume's pass-1 overview + v09-research/_SYNTHESIS.md + the relevant research-<angle>.md, producing one nano-detail document with concrete interfaces/DDL/protobuf/skeletons + Mermaid diagrams, citing sources, marking unproven facts UNVERIFIED. Each wave is committed + pushed; an adversarial review gates each volume. Wave order (resumed 2026-06-25): V2 data-plane → V3 control-plane → V4 clients → V10 design-system → V5 security (reviewed+reconciled→GO) → V6 deploy → V1 product → V7 execution → V8 testing → V0 meta .

Generation COMPLETE (2026-06-26): all 11 volumes (V0-V10) expanded — ~140 nano-detail docs, every .md carrying synced .html+.pdf siblings (§11.4.65, via scripts/testing/sync_all_markdown_exports.sh).

Outstanding quality items — updated 2026-07-04 (Rev 5 hardening pass):

- (a) **RESOLVED.** Adversarial reviews for V1/V6/V7/V8/V0 completed 2026-07-04 (independent gap-analysis + direct hardening, not a rubber-stamp — see Rev 5 changelog above for the specific defects found and fixed). All 11 volumes (V0-V10) are now independently reviewed at least once.
- (b) **STILL OPEN.** §11.4.168 — Mermaid fences render as raw source in HTML/PDF (needs a pandoc mermaid filter). This is a build/export-tooling gap, not a markdown-content gap, and is out of scope for a documentation hardening pass — tracked for whoever owns the docs_chain/pandoc export pipeline.
- (c) **RESOLVED (was already stale by 2026-07-04).** All 10 v09-research/research-<angle>.md per-angle dossiers exist on disk (confirmed by direct inspection during this pass), each with a Revision header (9 of 10 were missing the header and got it added in this pass) — this item's premise ("only wireguard split out") no longer matched the actual repository state.
- (d) **RESOLVED.** D-PKI-CA-TIER (two-tier issuing CA as MVP) is **operator-confirmed** per v00-meta/decision-register.md §4 (confirmed 2026-06-26, one day after this line was written —

this Outstanding-items list had simply not been updated since). No veto exercised; proceed with the two-tier design as binding for MVP.

- (e) **NEW, tracked for a future pass.** v03-control-plane/svc-coordinator.md §10.1 (added in this hardening pass) surfaces a genuinely open design question for Phase-2 multi-replica coordinators: fan-out vs. graph-apply consumption semantics for the shared event log. Not resolved here — correctly left open per the no-guessing discipline (needs a Phase-2 spike, not a documentation-pass decision).